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Studierichting: Informatica
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$$\int x^2 \cos^2 x dx = \int x^2 \cdot \frac{1+\cos 2x}{2} dx$$
$$= \frac{1}{2}\int x^2 dx + \frac{1}{2}\int x^2 \cos 2x dx$$
$$\left\{ \begin{array}{l} u = x^2 \\ dv = \cos 2x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 2x dx \\ v = \frac{1}{2}\sin 2x \end{array} \right.$$
$$= \frac{1}{2} \cdot \frac{x^3}{3} + \frac{1}{2}\left(\frac{x^2}{2}\sin 2x - \int x \sin 2x dx\right)$$
$$\left\{ \begin{array}{l} u = x \\ dv = \sin 2x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = dx \\ v = -\frac{1}{2}\cos 2x \end{array} \right.$$
$$= \frac{x^3}{6} + \frac{x^2}{4}\sin 2x - \frac{1}{2}\left(-\frac{x}{2}\cos 2x - \int -\frac{1}{2}\cos 2x dx\right)$$
$$= \frac{x^3}{6} + \frac{x^2}{4}\sin 2x - \frac{1}{2}\left(-\frac{x}{2}\cos 2x + \frac{1}{4}\sin 2x\right)$$
$$= \frac{x^3}{6} + \frac{x^2}{4}\sin 2x + \frac{x}{4}\cos 2x - \frac{1}{8}\sin 2x + C$$

References